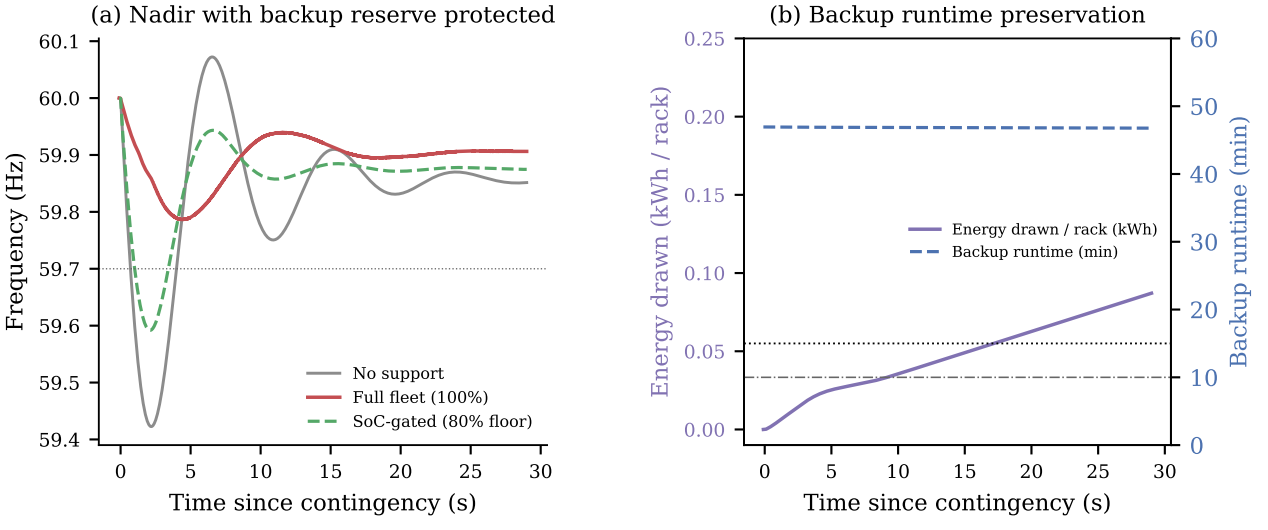
(a) Nadir with backup reserve protected
Frequency (Hz)
60.1
60.0
59.9
59.8
59.7
59.6
59.5
59.4
0
5
10
15
20
25
30
Time since contingency (s)
No support
Full fleet (100%)
SoC-gated (80% floor)
(b) Backup runtime preservation
Energy drawn (kWh / rack)
0.25
0.20
0.15
0.10
0.05
0.00
Backup runtime (min)
60
50
40
30
20
10
0
Energy drawn / rack (kWh)
Backup runtime (min)
0
5
10
15
20
25
30
Time since contingency (s)